Task-Technology Fit and Communication Synchronicity as Drivers of Microsoft Teams Collaboration Effectiveness: Evidence from Users in Bangladesh

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Abstract

Digital collaboration platforms combine task-support functionality with synchronous and asynchronous communication, yet these mechanisms are often examined separately. This study investigates whether task-technology fit (TTF) and communication synchronicity (CS) independently explain perceived Microsoft Teams collaboration effectiveness (CE), and whether CS strengthens the TTF-CE relationship. A cross-sectional survey dataset comprising 385 complete responses from Microsoft-platform users in Bangladesh was analysed. Data auditing confirmed complete focal responses, no duplicate respondent identifiers, valid 1-5 response coding, and no evidence of reverse-coded or miscoded focal items. Reliability was modest (Cronbach's α: TTF=.565, CS=.526, CE=.515), reflecting weak-to-moderate inter-item correlations rather than a single coding anomaly; deleting any one item did not materially resolve the problem. Composite-based diagnostics were more favorable (CR=.775/.760/.756; AVE=.535/.514/.509), while HTMT values remained below .85. Standardized composite-score regression showed positive associations of TTF (β=.257, p<.001) and CS (β=.211, p<.001) with CE, explaining 15.6% of its variance. The TTF×CS interaction was not significant (β=.016, p=.716). The findings support a parallel-mechanism interpretation: perceived task fit and communication-process support each matter for collaboration, but synchronicity does not amplify the contribution of task fit in this sample. The study contributes a boundary-condition test integrating Task-Technology Fit and Media Synchronicity Theory while explicitly recognizing measurement and cross-sectional limitations.

Keywords: Microsoft Teams; task-technology fit; media synchronicity theory; communication synchronicity; collaboration effectiveness; digital collaboration; Bangladesh

1. Introduction

Digital collaboration is now embedded in higher education and knowledge-intensive work. Microsoft Teams integrates meetings, chat, channels, file sharing, and Microsoft 365 resources, allowing users to move between real-time interaction and persistent asynchronous work [1], [2]. Yet feature availability does not itself guarantee effective collaboration. The same platform can support one task well and another poorly, and communication requirements vary across activities such as rapid problem solving, document review, coordination, and information exchange.

Task-Technology Fit (TTF) explains performance-relevant technology use through alignment between task requirements and technological capabilities [3], [4]. Media Synchronicity Theory (MST), by contrast, explains how media capabilities support conveyance and convergence processes and why different communication tasks require different temporal patterns [5]. Bringing these perspectives together creates a sharper question than whether users simply adopt or like Teams: do technological fit and communication-process support operate as independent drivers of collaboration effectiveness, or does communication synchronicity amplify the value of task fit?

This question is relevant in Bangladesh, where digital learning and work practices have expanded while connectivity, user capability, and organizational support remain heterogeneous [6]. Existing Microsoft Teams research documents collaborative learning and user experience [7]-[10], but platform-specific studies rarely test TTF and MST together as distinct explanatory mechanisms. The present study therefore focuses on three theoretically bounded constructs-TTF, CS, and CE-and tests both direct relationships and their proposed interaction.

The study addresses three gaps. First, it integrates a technology-task alignment perspective with a communication-process perspective without collapsing them into a generic technology-quality construct. Second, it moves beyond descriptive platform-use research by testing a boundary condition: whether synchronicity changes the strength of the TTF-CE relationship. Third, it provides primary evidence from Bangladesh while avoiding claims that a non-probability sample represents all national users. The theoretical contribution is therefore not merely contextual replication; it is an explicit test of whether TTF and MST operate additively or multiplicatively in perceived digital collaboration.

RQ1: To what extent is perceived task-technology fit associated with Microsoft Teams collaboration effectiveness?
RQ2: To what extent is communication synchronicity associated with collaboration effectiveness?
RQ3: Does communication synchronicity moderate the relationship between task-technology fit and collaboration effectiveness?

H1: Task-Technology Fit is positively associated with Microsoft Teams Collaboration Effectiveness.
H2: Communication Synchronicity is positively associated with Microsoft Teams Collaboration Effectiveness.
H3: Communication Synchronicity positively moderates the TTF-CE relationship, such that the association is stronger at higher levels of CS.

2. Theoretical Background and Hypothesis Development

TTF proposes that technology contributes to performance when its capabilities correspond to the requirements of the task [3]. Goodhue's measurement work further established fit as a user-evaluable property of information systems [4]. In collaborative environments, this alignment concerns whether platform functions are appropriate for coordination, information exchange, shared-document work, and

progress toward joint outputs. Group-support research similarly emphasizes joint consideration of task and technology characteristics [13], [14]. Meta-analytic and contemporary remote-work evidence continues to associate TTF with performance-relevant outcomes [15], [16]. Accordingly, users who perceive Teams as better aligned with their collaboration tasks should report greater collaboration effectiveness.

MST distinguishes conveyance, in which information is transmitted and processed, from convergence, in which participants develop shared meaning [5]. Rapid feedback may be valuable for convergence, whereas persistent and reprocessable communication can support conveyance. Teams provides both synchronous and asynchronous affordances. The CS construct used here captures timely communication, effective discussion, and support for immediate and delayed communication. Thus, higher CS should correspond with more effective coordination and information sharing [17]-[19].

TTF and MST describe different forms of alignment: TTF concerns fit between technology and task requirements, whereas MST concerns fit between media capabilities and communication processes. The stronger integrative proposition is that appropriate communication timing could enable users to convert a well-fitting feature set into coordinated action more effectively. H3 therefore tests a theoretically motivated interaction rather than assuming it. A null interaction would imply that the two mechanisms are better represented as parallel contributors in this context.

3. Method

This section explains the methodological decisions used to examine the proposed relationships among task-technology fit (TTF), communication synchronicity (CS), and collaboration effectiveness (CE). Because the credibility of the findings depends not only on statistical significance but also on transparency in data handling and measurement, the method proceeds from research design and participant characteristics to data-integrity checks, construct operationalization, reliability and coding diagnostics, analytical procedures, confirmatory model specification, and ethical/data-governance considerations. Particular attention is given to the limitations of the short three-item scales and to distinguishing directly reproducible composite-score statistics from latent-variable estimates that require dedicated PLS-SEM implementation. This approach provides an auditable empirical workflow while avoiding overstatement of measurement precision.

3.1 Research design and participants

The study uses a quantitative cross-sectional survey design. The unit of analysis is the individual user. The analytical dataset contains 385 complete focal responses from users in Bangladesh with relevant Microsoft-platform experience. Respondents include students, employed users, and self-employed/freelance users; accordingly, the revised title refers to users rather than university users. Because recruitment was purposive/criterion-based rather than probability-based, population representativeness is not claimed.

3.2 Data integrity audit

The workbook contains respondent IDs 1-400. IDs 1-385 contain complete focal responses, whereas rows 386-400 are blank templates. Among the 385 analytical cases, all nine focal items (TTF1-TTF3, CS1-CS3, CE1-CE3) are complete and all attention-check responses equal the expected value of 4. No duplicate respondent IDs were observed. Focal item values fall within the intended 1-5 scale. PAI1-PAI3 are present in the workbook but are not documented in the focal questionnaire/codebook and were therefore excluded from this article.

3.3 Measures

TTF was measured with three items assessing task fit, appropriateness of functions, and suitability for collaboration/productivity. CS was measured with three items assessing timely communication, discussion support, and support for immediate and delayed communication. CE was measured with three items assessing work coordination, information sharing, and progress on shared tasks. All items used a five-point agreement scale. The constructs are grounded in TTF and MST scholarship [3]-[5], [13], [14], while the exact short-form wording was tailored to the Microsoft collaboration context.

3.4 Reliability and coding diagnostics

A raw-data audit was undertaken before structural testing. The low alpha coefficients are reproduced directly from the 385 complete cases and are therefore not a transcription or calculation error. Inter-item correlations are positive but modest: TTF=.282-.315, CS=.231-.330, and CE=.197-.318. Corrected item-total correlations range from .366-.391 for TTF, .294-.368 for CS, and .287-.379 for CE. No focal item shows the negative correlation pattern expected from an uncorrected reverse-coded item. Moreover, deleting any single item fails to raise alpha to an acceptable level (maximum alpha-if-deleted=.479 for TTF, .495 for CS, and .482 for CE). The measurement weakness therefore reflects limited internal consistency of the short three-item scales rather than one obvious miscoded item.

3.5 Analytical strategy

To avoid presenting software-specific PLS-SEM statistics that were not generated by a validated PLS-SEM project, the confirmatory results reported in this article use transparent composite-score analysis. Each construct score is the arithmetic mean of its three items. TTF and CS are standardized before the interaction term is formed. Standardized ordinary least-squares regression estimates H1 and H2, and a product term estimates H3. Ten-thousand nonparametric bootstrap resamples are used to obtain confidence intervals for standardized coefficients. R^2 and local f^2 quantify explanatory and incremental contribution. Composite reliability, AVE, and HTMT are retained as diagnostic-not definitive-measurement evidence. This separation avoids conflating reproducible composite-score statistics with SmartPLS output.

3.6 PLS-SEM confirmatory model specification

The confirmatory PLS-SEM model is specified with three reflective Mode-A constructs (TTF, CS, and CE), each represented by its original three indicators; direct paths from TTF and CS to CE; and a two-stage latent-variable interaction representing TTF×CS→CE. Indicators are retained on theoretical grounds rather than deleted mechanically to improve reliability, because the item diagnostics do not identify a single anomalous indicator and post hoc deletion would narrow construct coverage. For a dedicated SmartPLS 4 replication of the model, bootstrapping with 10,000 subsamples and bias-corrected confidence intervals is appropriate. Full reporting should include outer loadings, Cronbach's alpha, rho_A, composite reliability, AVE, HTMT with inference, inner VIF, standardized path coefficients, R^2, f^2, Q^2, and PLSpredict where applicable [21]-[29]. Sample adequacy should be considered through power and effect-size logic rather than reliance on the 10-times rule.

3.7 Ethical and data-governance considerations

The study involved a non-interventional, minimal-risk questionnaire survey of adult users and collected no direct personal identifiers. Participation was voluntary, respondents were informed of the purpose of the study, and informed consent was obtained before participation. At the time the study was conducted, the authors' university did not require formal institutional ethics-committee approval for this type of non-clinical, non-interventional academic survey; consequently, no formal ethics approval number was

issued. The study was nevertheless conducted in accordance with generally accepted principles of voluntary participation, informed consent, confidentiality, and responsible handling of research data. The de-identified analytical data are retained by the research team and any sharing will remain subject to participant confidentiality and applicable institutional requirements.

4. Results

The results are reported in four stages so that the substantive findings can be interpreted against the characteristics and quality of the underlying data. First, the respondent profile is summarized to clarify the composition and scope of the analytical sample. Second, descriptive statistics and measurement diagnostics are presented for TTF, CS, and CE, including the mixed reliability evidence identified in the raw-data audit. Third, the hypothesized direct and moderating relationships are evaluated using standardized composite-score regression with bootstrap confidence intervals. Finally, sensitivity and interpretive checks are used to distinguish substantive conclusions from artifacts of coefficient scaling or construct overlap. Each table is introduced and interpreted in the text rather than being presented as a stand-alone statistical display.

4.1 Respondent profile

The 385-case sample is diverse in age, work status, education, location, and platform experience. The largest age group is 21-25 years (40.8%), followed by 18-20 (21.0%), 26-30 (17.4%), 31-40 (15.6%), and 41 or above (5.2%). Students account for 46.5% of respondents, employed users 35.3%, self-employed/freelancers 13.0%, and other statuses 5.2%. Dhaka contributes 32.7% of cases, with the remainder distributed across multiple locations.

The composition of the sample is important for interpreting the results. Nearly half of the respondents are students, but more than half are drawn from employed, self-employed/freelance, or other status categories. The evidence should therefore not be read as a student-only assessment of Microsoft Teams. At the same time, the concentration of respondents in younger age groups and the substantial Dhaka representation indicate that the sample reflects digitally active users more strongly than the full national population. These characteristics support the revised framing of the study as evidence from users in Bangladesh while reinforcing the decision not to claim national representativeness. The diversity in work status and location is nevertheless useful analytically because collaboration technologies are used across study, employment, and project-based work rather than within a single institutional setting.

4.2 Descriptive and measurement diagnostics

Table 1 summarizes the central tendency and dispersion of the three focal constructs together with the principal internal-consistency and convergent-validity diagnostics. Reporting these indicators together makes the central measurement issue transparent: respondents generally evaluated TTF, CS, and CE positively, but the short scales do not display equally strong evidence across all reliability criteria.

Table 1. Descriptive Statistics and Measurement-Model Diagnostics for the Focal Constructs

Construct	Mean	SD	Cronbach α	CR*	AVE*
TTF	3.987	0.523	0.565	0.775	0.535
CS	3.884	0.497	0.526	0.760	0.514
CE	3.923	0.505	0.515	0.756	0.509

**CR and AVE are composite-based diagnostic estimates from the observed item relationships; they are not represented as SmartPLS-exported statistics.*

The alpha values indicate limited internal consistency, while CR and AVE cross conventional composite-based thresholds. HTMT is .765 for TTF-CS, .639 for TTF-CE, and .609 for CS-CE, providing no indication of construct redundancy. Because the reliability evidence is mixed, structural findings are interpreted as association evidence from short composite measures rather than as definitive latent-variable estimates.

Taken together, the values in Table 1 suggest that the three constructs are usable for cautious association analysis but should not be presented as fully established latent scales. Mean scores are close to four on the five-point response scale, indicating broadly favorable perceptions of task fit, communication support, and collaboration outcomes. However, the relatively narrow standard deviations also indicate that responses are concentrated toward the positive end of the scale, which may restrict covariance and reduce the ability to detect more complex effects. The contrast between Cronbach's alpha and composite-based indicators is especially important. The CR and AVE estimates meet commonly used benchmarks, but alpha remains substantially below .70 for every construct. Because the raw-data audit shows that this pattern is not driven by a single reversed or miscoded item, the appropriate response is not mechanical item deletion but explicit qualification of the measurement evidence and prospective refinement of the scales in future research. The HTMT values below .85 provide a separate indication that TTF, CS, and CE are related without appearing empirically redundant.

4.3 Structural relationships

Table 2 reports the standardized coefficients for the two direct hypotheses and the proposed interaction. Standardization permits comparison of the relative magnitude of the associations on a common scale, while the bootstrap confidence intervals provide distribution-free uncertainty estimates. The table therefore distinguishes evidence for the two main-effect propositions from evidence for the stronger multiplicative claim embodied in H3.

Table 2. Standardized Structural Relationships and Hypothesis Tests

Hypothesis	Path	Standardized β	P	95% bootstrap CI	Decision
H1	TTF → CE	.257	<.001	[.159, .361]	Supported
H2	CS → CE	.211	<.001	[.099, .323]	Supported
H3	TTF × CS → CE	.016	.716	[-.072, .096]	Not supported

The direct-effects model explains 15.6% of the variance in CE (R^2=.156; adjusted R^2≈.151). TTF and CS are both positively associated with CE after accounting for each other. The interaction adds less than .001 to R^2 and is statistically non-significant. Thus, the data support parallel direct relationships rather than the hypothesized multiplicative synergy.

The coefficient for TTF is somewhat larger than that for CS, but the present design does not justify treating this difference as evidence that one mechanism is universally more important than the other. Both coefficients are modest in magnitude, and the model leaves most variance in CE unexplained. This is theoretically plausible because collaboration effectiveness can also depend on team norms, leadership, digital competence, task interdependence, system quality, information quality, prior experience, and organizational support. The principal result is therefore not that TTF and CS provide a comprehensive

explanation of collaboration effectiveness, but that each retains an independent positive association when considered simultaneously. In contrast, the confidence interval for H3 crosses zero and the interaction contributes virtually no incremental explained variance. The data thus provide no empirical basis for claiming that higher communication synchronicity strengthens the association between task fit and collaboration effectiveness in this sample.

4.4 Sensitivity and interpretation

Sensitivity checks were used to assess the robustness and interpretation of the standardized effects. Recalculation from the 385 complete cases yields standardized coefficients of .257 for TTF and .211 for CS. The moderate correlations among constructs are consistent with the HTMT evidence: the constructs share variance, as theory would predict, but are not so highly correlated that the direct paths merely reproduce the same underlying score. At the same time, the modest scale reliability means that coefficient magnitudes should not be overinterpreted. Measurement error generally attenuates observed relationships and can be especially consequential for interaction tests. For this reason, the null moderation finding is treated as an informative result in the present sample rather than definitive evidence that TTF and communication synchronicity can never interact.

The construct correlations are moderate (TTF-CS=.416, TTF-CE=.345, CS-CE=.318), which is consistent with related but non-redundant concepts. Both direct associations are positive and statistically significant, whereas the interaction is not. This pattern is stable across the principal composite-score checks and supports a cautious interpretation of TTF and CS as independent correlates of collaboration effectiveness.

5. Discussion

This section interprets the empirical findings in relation to Task-Technology Fit theory, Media Synchronicity Theory, and the wider literature on digitally mediated collaboration. The discussion is organized around the three hypotheses before considering the broader theoretical and practical implications. Particular emphasis is placed on the non-significant interaction because the absence of moderation is not treated simply as a failed hypothesis; rather, it is used to clarify the boundary of the proposed TTF-MST integration. The discussion also considers how measurement reliability, heterogeneous collaboration tasks, user diversity, and the Bangladesh context may shape the observed relationships, thereby separating what can be concluded from the present data from propositions that require future confirmatory research.

5.1 Task-technology fit and collaboration effectiveness

H1 is supported. Users who perceive stronger correspondence between Microsoft-platform capabilities and their collaboration tasks also report greater collaboration effectiveness. This pattern is consistent with the Technology-to-Performance Chain [3], later TTF measurement work [4], and research extending fit to collaborative and digitally mediated work [13]-[16]. The result reinforces a task-centered interpretation of collaboration technology: feature abundance matters less than whether the available functions correspond to what users need to accomplish.

The finding is also consistent with the broader logic of the Technology-to-Performance Chain: technology use is consequential when the characteristics of the technology correspond to the demands of the task rather than when adoption occurs in the abstract [3], [38]. In the present setting, Teams combines multiple capabilities, but users may experience those capabilities differently depending on whether they need to coordinate schedules, co-edit documents, exchange complex information, or

monitor progress. The positive TTF-CE association therefore shifts attention from platform possession to functional alignment. This distinction is important for both educational and work settings because organizations can deploy the same platform widely while users continue to experience uneven collaboration outcomes. Contemporary studies extending TTF to remote work and e-collaboration similarly suggest that fit remains relevant when work is digitally mediated [15], [16], [42]-[44]. The present evidence adds a platform-specific collaboration outcome while remaining cautious about causal language because task fit and perceived effectiveness were measured at the same point in time.

A further implication is that perceived fit may operate as a proximal evaluation of whether users can translate platform capabilities into task progress. The CE indicators concern coordination, information sharing, and progress on shared work; these are outcomes that require users to select and combine features rather than merely access them. Thus, the result is compatible with a mechanism in which fit reduces friction between what a collaborative task demands and what the digital environment enables. Future studies should test this mechanism with task-level designs in which the same user evaluates different collaboration activities, because aggregate user-level measures may conceal situations in which Teams fits one task well but another poorly.

5.2 Communication synchronicity and collaboration effectiveness

H2 is also supported. The CS items capture timely feedback, effective discussion, and the availability of immediate and delayed communication rather than assuming that maximum simultaneity is always desirable. The positive association therefore fits MST's distinction between communication processes and media capabilities [5], [17]-[19]. Teams can support rapid convergence through meetings and chat while preserving information for later processing through channels and files [1], [2].

This interpretation is more precise than equating synchronicity with continuous real-time communication. MST argues that communication performance depends on matching media capabilities to communication processes, particularly conveyance and convergence [5]. For information-rich conveyance, users may benefit from persistence, rehearsability, and reprocessability; for convergence, rapid feedback and parallel interaction may be more valuable. Teams is relevant because it permits movement between meetings, chat, persistent channels, files, and co-editing rather than forcing all collaboration into one temporal mode. The positive CS-CE relationship can therefore be interpreted as evidence that users value an environment in which communication timing can be adapted to the needs of the collaborative process.

The Bangladesh context gives this issue additional practical relevance without changing the theoretical claim. Connectivity and access conditions can make uninterrupted synchronous participation difficult for some users [6]. A collaboration environment that combines immediate interaction with persistent asynchronous alternatives may therefore support continuity when bandwidth, schedules, or work obligations constrain real-time participation. Nevertheless, the present CS measure is brief and does not separately operationalize the full set of MST capabilities. The result should consequently be read as support for perceived communication-process support, not as a complete empirical test of every dimension of MST.

5.3 The non-significant interaction as a boundary condition

H3 is not supported. The near-zero interaction suggests that perceived communication synchronicity does not materially change the TTF-CE relationship in this sample. This result is theoretically informative because it places a boundary on the proposed integration: task fit and communication-process support appear to operate as parallel mechanisms rather than mutually reinforcing ones. At least

three explanations merit future testing. First, the short CS measure may capture general communication support rather than the full conveyance-convergence distinctions of MST. Second, heterogeneous tasks may require different temporal configurations, attenuating an aggregate interaction. Third, limited measurement reliability reduces sensitivity to interaction effects, which generally require greater precision than main effects.

The null interaction deserves particular attention because the theoretical model deliberately tested whether integration of TTF and MST requires multiplicative synergy. The data indicate that it does not, at least with the present measures and sample. A user can perceive strong task fit even when communication timing is only moderate, and communication support can contribute to effective collaboration even when the platform is not perceived as optimally fitted to every task. This additive interpretation is theoretically meaningful: distinct mechanisms can coexist in an explanatory framework without one conditioning the effect of the other. Treating theoretical integration and statistical interaction as equivalent would impose a stronger claim than the evidence warrants.

Several methodological and substantive conditions may also suppress moderation. Interaction effects are more difficult to detect than main effects and are particularly sensitive to measurement error. The alpha coefficients in this study indicate that the short scales contain non-trivial measurement noise, which can attenuate a product-term relationship. In addition, the sample combines students, employees, freelancers, and other users whose collaboration tasks may differ in urgency, interdependence, equivocality, and duration. A single aggregate CS score may therefore average across qualitatively different communication requirements. Future work could model task interdependence and equivocality explicitly [44], separate conveyance from convergence, and test whether moderation emerges within more homogeneous task classes. Multi-group analysis may also reveal whether the interaction differs across user categories even though it is absent in the pooled sample.

Retaining the non-significant H3 preserves the confirmatory logic of the study. Rather than modifying the model after observing the data or deleting indicators to obtain a preferred result, the analysis treats the null interaction as evidence that narrows the theoretical claim and identifies a clear agenda for replication with stronger measures. This boundary-condition result is informative precisely because it distinguishes theoretical integration from an assumed requirement that every proposed relationship be statistically significant.

5.4 Theoretical implications

The study contributes by separating technology-task alignment from communication-process alignment and testing their joint role rather than assuming complementarity. The evidence supports a two-layer account of collaboration effectiveness: users benefit when the platform fits the work and when communication modes support appropriate timing. However, the absence of moderation indicates that theoretical integration need not imply statistical interaction. This distinction offers a testable boundary condition for future TTF-MST research.

First, the findings support conceptual separation between task-technology alignment and communication-process alignment. TTF explains whether the technological environment is suitable for the work to be performed, whereas MST explains how communication capabilities support the temporal and interpretive requirements of interaction. Their independent associations with CE indicate that these perspectives capture related but non-identical aspects of digital collaboration. This helps avoid reducing collaboration effectiveness to a generic attitude toward technology.

Second, the study refines the meaning of theoretical integration. Combining theories can involve complementary explanatory layers rather than a mandatory interaction effect. In the present model, TTF and CS jointly improve explanation of CE, but the contribution of one is not contingent on the level of the other. This suggests a parallel-mechanism architecture that future research can compare directly with mediation, moderation, or configurational alternatives. For example, communication processes might mediate some consequences of task fit, or different combinations of task characteristics and media capabilities might produce equivalent collaboration outcomes.

Third, the study highlights measurement as part of theory testing rather than a purely technical issue. The short scales provide adequate composite-based diagnostics but weak alpha reliability. Because interaction hypotheses require particularly precise measurement, future TTF-MST integration should use richer validated item sets that preserve construct breadth while improving internal consistency. A prospective design could distinguish technology characteristics, task characteristics, conveyance, convergence, and collaboration outcomes more explicitly. Such measurement refinement would allow researchers to determine whether the parallel-mechanism pattern is substantive or partly a consequence of attenuation in the present data.

Finally, the Bangladesh evidence broadens the contextual base of research that has often developed around Western organizational or educational settings. The study does not claim that national context itself causes the observed relationships, but it demonstrates that the core logic of task fit and communication-process support remains empirically relevant in a setting characterized by heterogeneous connectivity, institutional resources, and user roles. Comparative research can build on this by testing measurement invariance and structural differences across countries rather than assuming that digital collaboration mechanisms operate identically across contexts.

5.5 Practical implications

Institutions should evaluate collaboration platforms through task mapping rather than feature counting. Recurring activities such as group assignments, document review, project planning, feedback, and rapid problem solving can be matched to suitable Teams functions. Communication protocols should also distinguish urgency: synchronous meetings and rapid chat can support convergence-intensive work, while persistent channels and co-edited files can support information-heavy tasks that benefit from reprocessing. Training should therefore be scenario-based and should include low-bandwidth asynchronous alternatives where connectivity is constrained.

For universities, the findings imply that platform training should be organized around authentic collaborative tasks rather than menus of software features. Students and instructors can be shown how to select channels, meetings, chat, shared files, and co-editing workflows for different combinations of urgency, complexity, and need for shared understanding. For organizations, the same principle applies to project coordination, distributed teamwork, document review, and decision making. A task-mapping exercise can identify where the platform fits existing work and where process redesign or complementary tools may be required.

The positive CS association also argues against a one-size-fits-all communication policy. Requiring meetings for every issue can create coordination costs, while relying exclusively on asynchronous channels can slow convergence when rapid clarification is necessary. Managers and educators can instead establish communication norms that specify when synchronous interaction is expected, when asynchronous documentation is preferred, and how decisions made in meetings should be preserved for

later access. Such norms may be especially useful in settings where bandwidth and scheduling constraints vary across participants.

At the same time, the modest R^2 and measurement limitations caution against treating TTF and CS as a complete managerial diagnostic. Collaboration effectiveness also depends on human and organizational conditions that were not modelled here. Implementation decisions should therefore combine task and communication analysis with attention to digital literacy, accessibility, team norms, leadership, security, and institutional support. The findings identify two actionable dimensions of collaboration design, not a universal formula for successful digital teamwork.

6. Conclusion

This study examined whether TTF and communication synchronicity explain perceived Microsoft Teams collaboration effectiveness among users in Bangladesh. Both mechanisms show positive independent associations with CE, whereas their interaction is not supported. The findings therefore favor a parallel-mechanism interpretation over a synergy model. The contribution lies in testing-not assuming-the boundary between technology-task alignment and communication-process alignment in digital collaboration.

The study is cross-sectional, self-reported, and based on purposive sampling; causal inference and national representativeness are therefore not warranted. The most important measurement limitation is the low internal consistency of the three-item scales. Raw-data diagnostics show that this weakness is distributed across the short scales rather than attributable to one obvious coding error. Future research should use prospectively validated multi-item measures, document expert review and pilot testing, distinguish MST conveyance and convergence more explicitly, and collect task-level and behavioral indicators. Longitudinal, experimental, and multi-group designs could test whether the relationships vary by task interdependence, user status, or collaboration context. A confirmatory SmartPLS model should retain theoretically justified indicators and report full reliability, validity, predictive, and interaction diagnostics rather than optimizing the model through post hoc item deletion.

Declarations

Ethics and informed consent

The study used a non-interventional, minimal-risk questionnaire survey of adult participants. Participation was voluntary, the purpose of the study was communicated to respondents, and informed consent was obtained before participation. No direct personal identifiers were collected in the analytical dataset. At the time the research was conducted, the authors' university did not require formal institutional ethics-committee approval for this category of non-clinical, non-interventional academic survey; therefore, no formal ethics approval number was issued. The research was conducted with due regard for voluntary participation, informed consent, confidentiality, and responsible data handling.

Data availability

The de-identified analytical dataset is retained by the research team. Access or public sharing is subject to participant-consent and institutional requirements.

Conflict of interest

The authors declare no conflict of interest.

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Author contributions
Hasin Sadad Abir: conceptualization, data curation, investigation, formal analysis, visualization, writing-original draft, and writing-review and editing. Prof. Dr Kazi Abdul Mannan: supervision, conceptual guidance, methodology oversight, validation, interpretation, critical review, and writing-review and editing. Both authors reviewed and approved the final manuscript and accept responsibility for the integrity of their respective contributions.

Generative AI disclosure
Generative AI-assisted tools were used for language editing, manuscript-structure refinement, and analytical checking. All substantive interpretations, data decisions, references, and final claims were reviewed and remain the responsibility of the authors.

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Appendix A. Focal Questionnaire Items

TTF1: Microsoft tools fit the tasks I need to perform.

TTF2: Microsoft tools provide functions appropriate for my study/work tasks.

TTF3: Microsoft tools are well suited to my collaboration and productivity needs.

CS1: Microsoft Teams enables timely communication when rapid feedback is needed.

CS2: The communication features of Teams support effective discussion.

CS3: Teams provides suitable channels for both immediate and delayed communication.

CE1: Microsoft Teams helps my group coordinate work effectively.

CE2: Microsoft Teams improves information sharing among team members.

CE3: Microsoft Teams helps my team make progress on shared tasks.

Appendix B. Raw-Data Reliability Audit

Table 3. Raw-Data Reliability and Item-Diagnostic Audit

Construct	Inter-item r range	Corrected item-total r range	Maximum α if one item deleted	Interpretation
TTF	.282-.315	.366-.391	.479	No single problematic item
CS	.231-.330	.294-.368	.495	CS3 weakest, but deletion insufficient
CE	.197-.318	.287-.379	.482	CE1 weakest, but deletion insufficient

All nine focal variables use the intended 1-5 direction. No negative inter-item correlation indicates an uncorrected reverse-coded item. Accordingly, post hoc deletion solely to raise reliability is not recommended.

Appendix C. SmartPLS 4 Confirmatory Model Specification

Measurement model: TTF, CS, and CE specified as reflective Mode-A constructs with their original three indicators. Structural model: TTF→CE; CS→CE; TTF×CS→CE. Moderation: two-stage approach to reduce unnecessary indicator proliferation with short reflective scales. Bootstrapping: 10,000 subsamples; report bias-corrected 95% confidence intervals. Measurement reporting: outer loadings, α, rho_A, CR, AVE, HTMT and bootstrapped HTMT inference. Structural reporting: inner VIF, standardized paths, R^2/adjusted R^2, f^2, Q^2 and PLSpredict where justified. Do not delete indicators merely to manufacture $\alpha \geq .70$; any deletion must be theoretically defensible and prospectively validated in a new sample.